

Data and infrastructure

Contribution 1

GLES panel data

Linked respondent-wave records

Climate-growth and left-right

Grouped and original response scales

Evaluation design

4 tasks and 4 prompt conditions

1

Raw GLEs files

Selected waves
and variables

2

Harmonized records

Respondent IDs
and profiles

3

Condition-specific prompts

Profile and question
Prompt information

4

External generation

LLaMA-Factory
By configuration

5

Linked evaluation records

Predictions + answers
Parsing and retention

Respondent and wave links connect prompts, outputs, observed answers, and parser outcomes.

Mistral-7B • Qwen2.5-7B (4-bit) • Qwen2.5-32B • Qwen2.5-72B • Llama-3.3-70B

Evaluation of current states and observed transitions

Contribution 2

RQ1 Current-state accuracy

Comparison

No-time LLM vs.
covariates-only ordinal probit

Main metric

Exact-match accuracy

Supplementary comparisons

Other non-trajectory prompts
original-scale distance metrics

RQ2 Prior-state information

Prompt comparison

Trajectory vs. non-trajectory
prompts on common records

Baseline comparison

Modal rule, carry-forward, and
lag-and-covariates ordinal probit

Metrics

Exact-match accuracy and gaps
prior-state agreement

RQ3 Observed transitions

Transition decomposition

Stable and changing accuracy
contributions to pooled accuracy

Change recovery

Departure → correct direction
→ exact current response

Diagnostics

False persistence
anchored-change heatmaps

Supporting diagnostics

Distributional fidelity

Total variation distance

Representational fidelity

Intersectional subgroup
accuracy

Structural fidelity

Coefficient alignment

Self-trajectory change

Changes in successive
predictions